Understanding and Improving Compositionality of Stable Diffusion

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Abstract

Modern text-to-image diffusion models struggle with compositional reasoning because standard text encoders lack strict spatial grounding. In this work, we investigate whether targeted text-encoder interventions can resolve these spatial deficits without the prohibitive computational cost of full-model retraining. We perform a modular "weight surgery" on Stable Diffusion 3.5 Large Turbo (SD3.5), replacing its default CLIP-L encoder with two compositionally-aware variants: CLIC RedCaps and CLIC CogVLM. Evaluating across a standardized benchmark of over 1,300 relational prompts using Qwen3-VL-30B and manual annotation, we demonstrate some improvements in spatial accuracy; particularly in complex directional and containment tasks. On the GenEval2 Soft-TIFA benchmark (800 prompts), both CLIC variants improve over the Turbo baseline, with CogVLM reaching the highest atom-level score (56.52% AM vs. 55.48% baseline) and RedCaps the highest prompt-level consistency (10.68% GM vs. 8.91% baseline). Furthermore, our error analysis uncovers a "Reasoning-Output Paradox" within automated VLM evaluators and highlights the persistent dominance of semantic priors over structural nouns. Ultimately, our findings establish that visual compositionality relies heavily on text-encoder grounding rather than mere model scale, providing a highly efficient framework to enhance spatial reliability in production-ready diffusion pipelines.

1. Introduction

1.1. Problem Motivation

Modern text-to-image diffusion models have achieved remarkable success in generating high-quality, photorealistic images (Dhariwal & Nichol, 2021; Lugmayr et al., 2022; Poole et al., 2022; Saharia et al., 2022). However, they continue to struggle with compositional understanding (Chatterjee et al., 2024; Adamkiewicz et al., 2026), defined as the ability to correctly interpret and render spatial relationships, object interactions, and logical constraints specified in natural language prompts. While models such as Stable Diffusion 3.5 (SD3.5) (Esser et al., 2024; Rombach et al., 2022) can generate stunningly detailed images of individual objects, they systematically fail when prompts require precise spatial positioning, logical negation (Ban et al., 2024), or complex relational reasoning. This limitation represents a critical gap between language comprehension and visual generation, ultimately restricting the practical utility of these models for tasks that demand rigorous spatial control.

This research problem fundamentally stems from the nature of the training data used for standard Vision-Language Models (VLMs) (Bordes et al., 2024). Foundational models like CLIP (Radford et al., 2021) are frequently trained on vast datasets of loose, web-scraped image-caption pairs where strict spatial precision is rarely a priority. Historically, prior work has primarily focused on scaling diffusion architectures and optimizing general image quality metrics such as Importance Sampling (Grover et al., 2019) and standard CLIP scores (Hessel et al., 2022), while the systematic investigation of compositional failures has remained largely limited. Because most evaluation studies verify mere object presence rather than analyzing accurate spatial relationships, fundamental limitations in language-to-visual grounding are often masked. Consequently, the text embeddings fed into the diffusion transformer may inherently lack the explicit "spatial coordinates" necessary for the denoising process to arrange generated objects correctly.

Understanding and addressing compositional failures in diffusion models has several important implications:

1.1.1. Scientific Value

Compositionality is a foundational property of both human language and cognitive processing. Investigating precisely why state-of-the-art vision-language models fail at seemingly simple spatial reasoning tasks—such as accurately positioning a bed above a book—provides critical insights into the inherent limitations of current training paradigms

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and model architectures.

1.1.2. Practical Value

Downstream applications requiring rigorous spatial precision (Ruiz et al., 2023; Tian et al., 2023; Hammoud et al., 2024; Voetman et al., 2023) are currently severely bottlenecked by the inability of underlying models to reliably interpret spatial language. Consequently, improving a model's compositional reasoning capabilities would significantly expand the reliability and utility of text-to-image systems, particularly within professional creative workflows and controlled generation pipelines.

1.2. Architectural Insight

By systematically isolating and replacing specific model components (namely, the text encoders) and comprehensively measuring the resulting impact, this work contributes to a deeper understanding of which architectural elements are primarily responsible for compositional understanding as opposed to general image quality. Crucially, our research focuses on determining whether specifically targeting the text encoding stage can resolve these spatial reasoning deficits without incurring the prohibitive computational costs of full model retraining.

1.3. Research Aim and Overview

In this work, we systematically quantify the failure rates across a diverse set of spatial and relational concepts within SD3.5. We specifically investigate whether replacing SD3.5's default CLIP-L encoder with CLIC (Compositional Language-Image Contrastive) (Peleg et al., 2025), an encoder explicitly trained on compositionally dense image-text pairs, measurably enhances the model's overall compositional reasoning capabilities.

To this end, we detail a comprehensive technical methodology for replacing the model's internal text encoder with two specialized variants: CLIC RedCaps[1], trained on 12 million Reddit image-text pairs rich in detailed spatial descriptions (Desai et al., 2021), and CLIC CogVLM[2], which is aligned with high-performance visual features (Wang et al., 2024). We analyze model performance across 13 distinct relational phrases (above, below, to the left of, to the right of, between, behind, closeby, far from, in front of, on top of, under, over, and without) by utilizing automated VLM-based evaluation via Qwen3-VL (Bai et al., 2023). We further validate our findings on the GenEval2 Soft-TIFA benchmark (Kamath et al., 2025) to situate our results within a standardized compositional evaluation framework. In addition to our empirical findings, we develop an automated, VLM-based evaluation infrastructure designed to assess these spatial relationships at scale, and we document the complete pipeline required for converting and integrating OpenCLIP models into production-level diffusion pipelines.

The remainder of this report is organized as follows: Section 2 details our data collection processes and architectural methodology; Section 3 presents both the quantitative and qualitative results of our experiments; and Section 4 discusses our core findings, acknowledges current limitations, and outlines directions for future work.

2. Methods and Data

The methodology for this research was structured as a multi-phase experimental pipeline designed to diagnose, address, and evaluate compositional reasoning deficits in state-of-the-art diffusion models. The following sections detail the iterative process of model selection, prompt engineering, architectural modification, and multi-modal evaluation.

2.1. Model Selection and Baseline Configuration

The study began with an extensive benchmarking phase to identify a robust foundation for compositional analysis. After evaluating several iterations of the Stable Diffusion family, Stable Diffusion 3.5 (SD3.5) Large Turbo[3] was selected due to its sophisticated triple-text-encoder architecture and high-fidelity "Turbo" distillation.

2.1.1. Deterministic Controls

To ensure experimental reproducibility and eliminate stochastic variance during comparative analysis, all image generations were locked to a fixed random seed of 42.

2.1.2. Hyperparameter Settings

Generations utilized a guidance scale (Zhan et al., 2026) of 4.0 and 28 inference steps, adhering strictly to the optimized defaults for the SD3.5 Turbo pipeline.

2.1.3. Output Format

Generated images were stored as raw PyTorch tensors (`float32`) and systematically normalized to ensure compatibility with downstream Vision-Language Model (VLM) assessment frameworks.

[1] https://huggingface.co/nmndeep/CLIC-ViT-L-14-224-PixPr-RedCaps

[2] https://huggingface.co/nmndeep/CLIC-ViT-L-14-224-CogVLM

[3] https://huggingface.co/stabilityai/stable-diffusion-3.5-large-turbo/blob/main/LICENSE.md

2.2. Exploratory Pilot and Problem Identification

Before formalizing the evaluation dataset, an unconstrained exploratory phase was conducted to map the specific failure modes of the baseline SD3.5 model.

1. **Baseline Diagnostics:** Initial trials confirmed that while the model achieved exceptional photorealism, it suffered from obvious compositional failures. These were most notable in numerical reasoning (counting) (Li et al., 2025; Zhao et al., 2025) and in processing logical negations such as the phrase "not" (Ban et al., 2024).
2. **Spatial Blind Spots:** Systematic testing revealed persistent "blind spots" in basic vertical and horizontal stratification. For instance, prompts requiring objects to be "above" or "to the left of" one another frequently resulted in inverted coordinates or side-by-side placements.
3. **Relational Complexity:** We identified that SD3.5 struggled significantly with exclusion (e.g., "without") and physical interactions (e.g., "holding"), where objects were often inappropriately merged or rendered in complete isolation.

2.3. Dataset Construction: Self-Bench and Standardized Prompting

To explicitly isolate the linguistic understanding of the text encoder from the generative capacity of the diffusion transformer, we utilized the Self-Bench (Jeong et al., 2025) framework.

1. **Class Extraction:** We extracted 100 unique class pairs (Appendix 5.1), such that they are visually distinct object classes (e.g., "giraffe", "book", "broccoli") and to ensure that any failure in the final image could be confidently attributed to a misunderstanding of the spatial relationship, rather than an inability to render the objects themselves.
2. **Prompt Taxonomy:** A comprehensive benchmark of over 1,300 unique prompts was constructed across 13 core spatial and relational phrases: above, below, to the left of, to the right of, between, behind, closeby, far from, in front of, on top of, under, over, and without.
3. **Template Structure:** Every prompt followed a rigid syntactic template: "a photo of a [class1] [phrase] a [class2]". For the phrase "between", we adapted the template to: "a photo of a [class1] between [class2]s". This strict standardization allowed for controlled, apples-to-apples comparisons across the baseline and custom model variants.

We also generated prompts and images using single classes (or two classes) with verbs like "running", "standing", "sitting", "holding", "not holding", "touching", and "not touching" to further probe the model's understanding of physical interactions and exclusion. Further we also explored prompts with simple negation phrases like "without" and "and not", which showed typical failure modes where the model either ignored the negation and generated both objects. These findings highlighted the model's struggle with logical negation and exclusion, which is a critical aspect of compositional reasoning. These prompts were not included in the final evaluation benchmark, but they provided critical insights during the exploratory phase and informed our understanding of the model's relational reasoning capabilities.

2.4. Architectural Intervention: Encoder Replacement Pipeline

The central hypothesis of this work is that replacing the primary CLIP-L text encoder (Text Encoder 1) with a compositionally aware alternative can significantly improve the model's understanding of spatial semantics. We implemented a modular replacement of this component using CLIC (Compositional Language-Image Contrastive) variants, as depicted in Figure 1.

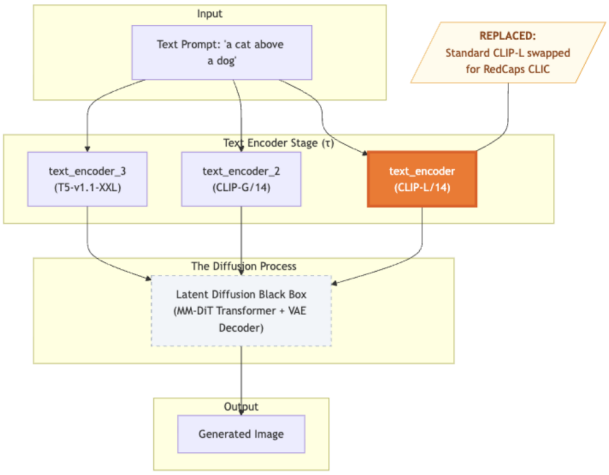


Figure 1. Architectural overview of the CLIP-L replacement with CLIC.

2.4.1. CLIC Variants: RedCaps and CogVLM

We integrated two distinct text encoders inherently optimized for compositional awareness:

1. **RedCaps CLIC:** Trained on 12 million image-text pairs sourced from Reddit, where alt-text frequently features highly descriptive spatial arrangements in-

tended for accessibility.

2. **CogVLM CLIC:** A variant aligned with the CogVLM visual feature space, specifically designed to enhance the bridge between generative modeling and discriminative spatial understanding.

2.4.2. Technical "Weight Surgery"

Because the CLIC weights were distributed in the OpenCLIP format, they were architecturally incompatible out-of-the-box with the HuggingFace-based SD3.5 pipeline. To resolve this, we developed a custom conversion utility to perform the following transformations:

1. **QKV Matrix Decomposition:** The fused `in_proj_weight` matrix of shape 2304×768 was mathematically split into three separate Query (Q), Key (K), and Value (V) projections, each possessing a shape of 768×768.

2. **Activation Alignment:** The model configuration was explicitly updated to use the QuickGELU activation function, defined as $x \cdot \text{sigmoid}(1.702 \cdot x)$. We empirically observed that failure to align with QuickGELU led to immediate latent distribution collapse during generation.

3. **Dimension and Layer Mapping:** Weights were systematically mapped from OpenCLIP's `resblocks` nomenclature to HuggingFace's `encoder.layers` format, which included performing a necessary transpose operation on the final `text_projection` matrix.

2.4.3. Token Length

To satisfy standard Stable Diffusion architectural requirements, the `CLIPTokenizer` was strictly padded or truncated to a maximum token length of 77.

2.5. Multi-Prompt Evaluation

To analyze the resulting dataset of over 3,000 generated images, we initially evaluated both LLaVa (Liu et al., 2023) and Qwen3-VL. Our pilot analysis revealed that Qwen3-VL outputs aligned much more closely with manual human annotations compared to LLaVa (Chen et al., 2025) across a critical subset of phrases ("above", "below", "to the left of", "to the right of", and "between"). Consequently, we utilized the Qwen3-VL-30B-A3B-Instruct model[4] for the primary evaluation. To ensure robust assessment, we employed three distinct prompting strategies (Liu & Chilton, 2023) for the VLM evaluator:

1. **Direct Query:** A simple binary question regarding the spatial relationship, formatted as: *"Is the {class1} {phrase} the {class2}? Answer Yes, No, or Ambiguous."*

2. **Context-Prime (Expert Role):** Leveraging a system prompt with established context (Zhang et al., 2024): *"You are an Expert Visual Analyst and expert at spatial reasoning. 1. Identify the {class1} and describe its position. 2. Identify the {class2} and describe its position. 3. Based on their relative coordinates, is the {class1} {phrase} the {class2}? Conclusion: [Yes/No/Ambiguous]."* This multi-step mode successfully generated reasoning traces that provided deeper insights into VLM failure modes. For the phrase "between", the prompt was modified to: *"You are an expert at spatial reasoning. Task: Is the {class1} between the {class2}s? Step 1: Identify and locate the {class1}. Step 2: Identify the group of {class2}s or the boundary objects. Step 3: Analyze if the {class1} is positioned in the intermediate space separating the {class2}s. Conclusion: [Yes/No/Ambiguous]."*

3. **Inverse Logic:** Prompting the VLM to describe the relationship present in the image to cross-reference against the original generative prompt: *"You are an Expert Visual Analyst and expert at spatial reasoning. 1. Is the {class2} {inverse_phrase} the {class1}? 2. If so, then the {class1} is {phrase} the {class2}. Conclusion: [Yes/No/Ambiguous]."*

Within our evaluation framework, a generated image is categorized as "Ambiguous" when the spatial relationship between the prompted objects is visually confusing or indeterminate, preventing a strict "Correct" or "Incorrect" classification. In these cases, the objects themselves may be identifiable, but their physical arrangement, perspective, or interaction within the scene makes the targeted spatial preposition impossible to evaluate definitively.

Manual Annotation: To ground the automated metrics and ensure VLM reliability, a representative subset of 10 to 20 images per phrase - model combination was sampled for manual review. Each image was assigned one of three labels: Correct, Incorrect, or Ambiguous; which were then compared against the Qwen3-VL predictions to measure inter-evaluator agreement.

2.5.1. Majority Vote Consensus Mechanism

To consolidate the multi-modal assessment and calculate the final accuracies reported in our quantitative results, we implemented a Majority Vote Consensus mechanism to aggregate the VLM responses. For each generated image, Qwen3-VL-30B produced categorical predictions ("Correct", "Incorrect", or "Ambiguous") across the applied prompting

[4] https://huggingface.co/Qwen/Qwen3-VL-30B-A3B-Instruct

styles.

Most spatial phrases were evaluated using all three styles (Direct Query, Expert, and Inverse Logic). However, certain phrases were evaluated using only two styles (Direct Query and Expert) due to linguistic constraints preventing natural inverse phrasing. The final classification for each image was determined by calculating the strict majority among these predictions. For example, if three prompting styles yielded [Correct, Correct, Incorrect], the final label was recorded as Correct.

To ensure rigorous benchmarking, we instituted a conservative tie-breaking protocol: in any instance lacking a clear majority—such as a three way split of [Correct, Incorrect, Ambiguous] across three prompts, or a conflicting [Correct, Incorrect] split for phrases with only two prompts the final prediction automatically defaulted to "Ambiguous." This guarantees that the models are only credited with a correct generation when the automated evaluator demonstrates consistent, high-confidence agreement.

2.6. GenEval2 Soft-TIFA Evaluation

We evaluated all three model variants on GenEval2 (Kamath et al., 2025), an 800-prompt benchmark measuring Text-to-Image faithfulness across primitive visual concepts (objects, attributes, counting, spatial relations, and verbs) with varying degrees of compositionality.

Evaluation used the benchmark's recommended *Soft-TIFA* method: for each prompt, a VQA oracle (Qwen3-VL-8B-Instruct) answers a fixed set of question–answer pairs about the generated image, producing a soft probability per question. We report two aggregate scores: the arithmetic mean (AM) of per-prompt scores, which estimates atom-level accuracy, and the geometric mean (GM), which estimates prompt-level accuracy and penalises any single failed atom more heavily. All three variants were evaluated on images generated with identical settings (28 inference steps, guidance scale 4.0, seed 42).

2.7. Hardware and Computational Environment

All experiments were executed on the University of Tübingen's high-performance cluster, utilizing NVIDIA A100 (40GB) GPUs. The software environment was built utilizing `diffusers` version 0.24.0 and `transformers` version 4.35.0. The total computational footprint including exploratory pilots, weight conversion, large-scale generation, and VLM inference amounted to approximately 150 GPU hours.

3. Results

In this section, we present the quantitative and qualitative outcomes of our experiments. We compare the baseline Stable Diffusion 3.5 Large Turbo (SD3.5) model against the CLIC RedCaps and CLIC CogVLM variants across a comprehensive set of spatial and relational benchmarks. Our analysis utilizes a multi-modal approach, combining large-scale automated VLM assessment, granular manual annotation to ensure evaluator reliability, and GenEval2 Soft-TIFA benchmarking to contextualize our findings within a broader compositionality framework.

3.1. Exploratory Pilot Findings

Prior to the structured benchmarking phase, our unconstrained pilot exploration revealed several notable failure modes inherent to the baseline SD3.5 model. Examples of these anomalies are illustrated in Figure 2.

1. **Semantic Prior Dominance:** The model frequently prioritized high-probability object tokens over structural or contextual nouns. For example, generating a literal pizza instead of interpreting the full concept of a "pizza flyer".

2. **Implicit Geometric Constraints:** Prompts involving complex structural objects, such as a "chessboard", consistently failed to adhere to the implicit 8×8 grid logic, resulting in irregular, non-Euclidean patterns.

3. **Action-Location Confusion:** In compositional prompts such as "A sketching X at Y", the model often rendered "A at X", "A at Y", or fundamentally confused the subject being sketched (X) with the physical location (Y) of the sketching action.

4. **Physical Interaction Failures:** Biological interactions, specifically intra-species or inter-species interactions, such as a "baby cow drinking milk", frequently resulted in fused anatomies or physically impossible object orientations.

5. **Optical Physics and Reflection:** The model struggled significantly with "half-reflection" tasks, demonstrating a lack of foundational understanding regarding the optical properties of mirrors and reflection.

3.2. Quantitative Analysis of Text Encoder Weights

Analysis of the surgical replacement revealed significant statistical shifts within the model's embedding space. The weights of the RedCaps and CogVLM text encoders demonstrated variances of 1.395×10^{-3} and 2.102×10^{-3} compared to the original CLIP-L encoder, respectively. This



(a) "A pizza place flyer floating on water." (b) "A chessboard carved into stone." (c) "A person sketching a skyline on rooftop." (d) "A mirror reflecting only half of a person's face." (e) "A baby cow drinking milk from mother cow's udder."

Figure 2. Notable failure modes discovered during the unconstrained exploration phase.

confirmed the successful integration and functional divergence of the replacement text encoders within our pipeline.

Furthermore, we performed a layer-by-layer weight divergence analysis between the original CLIP-L and the custom CLIC variants. Frobenius norms across the 12 transformer layers showed a consistently increasing trend from layer 0 to layer 11, with layer 11 exhibiting the largest total weight magnitude in both variants. Within the attention mechanisms, all projection matrices (Query, Key, Value, and Output) showed elevated norms in CogVLM relative to RedCaps, with Key projections (1.21× higher) and Value projections (1.22× higher) exhibiting the most pronounced differences. These elevations in attention projections suggest that the CogVLM variant may employ more fine-grained token relationships compared to the standard model. More detailed values across the layers of these weight differences is provided in Appendix 5.2.

3.3. Comparative Model Behavior and Manual Observations

Manual validation of 10 to 20 images per concept revealed distinctly different behavioral patterns between the standard baseline and the custom models.The aggregated results of this manual annotation are presented in Table 1 and the labelling is done as explained in the Methods section.

3.3.1. Baseline SD3.5

From our investigation, we identified the following persistent limitations in the baseline model, as visualized in Figure 3:

1. **Positional Inversion and Order Bias:** The baseline performs poorly on the relation "above," but fails significantly more often on "below," as it frequently defaults to placing the first-mentioned object in the prompt above the second.

2. **Lack of Scene Diversity:** The model is largely unable to generate counter-intuitive scenes rarely found in natural training datasets, such as a "ball above book" or a "bench above cow".

3. **Directional Disparity:** While the model demonstrates adequate performance for "to the left of", it suffers a notable performance degradation when processing the inverse relation, "to the right of".

4. **Inverse Spatial Logic:** For spatial prepositions like "over" and "under", the model frequently generates objects exhibiting the exact inverse of the requested spatial relationship.

3.3.2. CLIC RedCaps and CogVLM

Compared to Standard SD3.5, we observed the following improvements and anomalies, with examples shown in Figure 4:

1. **Improved Verticality and Containment:** RedCaps significantly outperforms the baseline in vertical reasoning and containment tasks (e.g., "under", "over", "between"), although a minor fraction of images still exhibit generation errors.

2. **Reduction in Ambiguity:** Both custom variants produced markedly fewer ambiguous or entirely fused images compared to the standard model.

3. **CogVLM Performance Anomalies:** CogVLM behaves similarly to RedCaps overall but surprisingly performs worse on the phrases "to the right of" and "between" compared to RedCaps, despite still exceeding the baseline's performance. In certain edge cases, it replicates the exact spatial failures seen in Standard SD3.5.



(a) "A photo of a giraffe below a hair drier." (b) "A photo of a bench above a cow." (c) "A photo of a book above a sports ball." (d) "A photo of a toilet to the left of a potted plant." (e) "A photo of a toilet to the right of a potted plant."

Figure 3. Standard SD3.5 outputs demonstrating spatial failure modes.



(a) "A photo of a giraffe below a hair drier." (b) "A photo of a bench above a cow" (c) "A photo of a book above a sports ball." (d) "A photo of a toilet to the left of a potted plant." (e) "A photo of a toilet to the right of a potted plant."

(f) "A photo of a giraffe below a hair drier." (g) "A photo of a bench above a cow" (h) "A photo of a book above a sports ball." (i) "A photo of a toilet to the left of a potted plant." (j) "A photo of a toilet to the right of a potted plant."

Figure 4. RedCaps CLIC (top row) and CogVLM CLIC (bottom row) variant outputs.

3.4. Quantitative Benchmark Results

Following the pilot, we utilized a standardized taxonomy derived from the Self-Bench classes to evaluate over 1,300 generated prompts. Final accuracy was determined through automated Qwen3-VL-30B scoring, cross-referenced against our manual annotations. The comprehensive results are summarized in Table 2.

3.5. Automated Evaluator Validation

3.5.1. LLaVA

Our initial automated evaluation across core spatial phrases (such as "above", "below", "to the left of", and "to the right of") revealed that LLaVA yielded significant inconsistencies, achieving an overall accuracy of just 56.4% (F1: 0.26). Specifically, on the phrase "above", LLaVA achieved a mere 43.18% accuracy. Deeming this performance insufficient for reliable benchmarking, we pivoted to Qwen3-VL. This adjustment resulted in substantial improvements: our direct Qwen query approach achieved 73.48% overall accuracy (F1: 0.62), while the expert reasoning-based Qwen approach reached a comparable 71.46% accuracy (F1: 0.62).

3.5.2. Qwen3-VL-30B

We carefully measured the agreement between the Qwen3-VL-30B predictions and human evaluation. Table 3 presents a sample confusion matrix for the RedCaps SD variant (Phrase: "Above", $n = 20$), demonstrating high evaluator reliability. Table 4 illustrates the overall agreement between manual labels and Qwen3-VL accuracies across different

Table 1. Manual annotation results across the different SD3.5 variants, formatted as (Ambiguous, Incorrect, Correct). The number of manually checked images is listed in parentheses next to the phrase.

Phrase	Standard	RedCaps	CogVLM
above (20)	(4,7,9)	(1,5,14)	(1,6,13)
below (20)	(4,8,8)	(1,12,7)	(0,9,11)
to the left of (20)	(4,5,11)	(0,9,11)	(0,8,12)
to the right of (20)	(1,17,2)	(0,9,11)	(0,11,9)
between (20)	(0,7,13)	(3,2,15)	(0,5,15)
behind (10)	(0,6,4)	(1,7,2)	(1,7,2)
closeby (10)	-	(0,1,9)	(0,1,9)
far from (10)	-	(0,7,3)	(0,6,4)
in front of (10)	(0,4,6)	(0,4,6)	(0,3,7)
without (10)	(0,7,3)	(0,6,4)	(1,5,4)

Table 2. Comparative accuracy across SD3.5 model variants using Majority-Voted VLM Scoring.

Phrase	Standard SD3.5	RedCaps SD	CogVLM SD
above	72.73	85.00	90.00
below	70.00	75.00	90.00
to the left of	78.79	80.00	75.00
to the right of	84.85	65.00	80.00
between	68.00	-	55.00
behind	63.16	60.00	60.00
in front of	68.42	50.00	70.00
on top of	33.33	30.00	20.00
over	70.00	40.00	40.00
under	40.00	40.00	30.00
without	70.00	30.00	30.00

relational phrases.

Table 3. Sample Confusion Matrix demonstrating the high alignment (+80% match) for the RedCaps model evaluating the "above" phrase utilizing the "expert reasoning" prompt strategy.

	QWEN Yes	QWEN No	QWEN Amb
Manual Yes	15	2	0
Manual No	0	3	0
Manual Amb	0	0	0

3.5.3. VLM Logic and Reasoning Patterns

Our deeper analysis of Qwen3-VL's internal reasoning traces identified critical failure bottlenecks in the evaluation pipeline. The VLM often reasons correctly about the visual elements but concludes with the incorrect logical answer. For example, regarding the image referenced in Figure 5, generated by the prompt "A photo of a bed under a book", Qwen3-VL correctly identifies both objects and describes their relative positions accurately:



Figure 5. An example of a Qwen3-VL failure mode where the model correctly identifies the objects and their relative physical positions but ultimately formulates an incorrect spatial conclusion.

"1. The bed is identified as the large surface covered with teal bedding, including sheets and pillows, located in the lower and middle portions of the image... 2. The book is an open book with visible text, resting on top of the bed... 3. The book is physically located on top of the bed, not above it in a vertical sense. Therefore, the bed is not under the book in a spatial sense; rather, the book is on the bed. Conclusion: No."

Despite this accurate breakdown, it ultimately concludes that the bed is *not* under the book: a clear failure in spatial reasoning logic. This contradictory pattern was observed across multiple phrases. Furthermore, similar to the diffusion models themselves, the VLM evaluator frequently conflates "left" and "right". For the phrase "in front of", the VLM often rigidly expects direct physical occlusion, whereas our manual annotation accepted lateral foreground positioning.

Additionally, during inverse prompting modes, the VLM frequently produced excessively verbose reasoning outputs. When the final answer was cut off due to token limits, this resulted in an automatic "Ambiguous" label across all models, though this specific failure occurred most frequently when evaluating the standard SD outputs.

3.6. GenEval2 Soft-TIFA Benchmark

To complement our spatial-relation evaluation with a broader, question-answering-based compositional metric, we evaluated all three variants on the GenEval2 benchmark (Kamath et al., 2025) using Qwen3-VL-8B-Instruct as the oracle. We report the arithmetic mean (AM, atom-level) and geometric mean (GM, prompt-level) of per-prompt Soft-TIFA scores in Table 5.

The 14 pp gap in AM between SD3.5-Large and our Turbo baseline is consistent with the known quality-speed trade-off of distillation. The GM gap is even starker (22.80 vs.

Table 4. Qwen3-VL Agreement with Manual Annotations

Agreement Level	Standard SD3.5	RedCaps SD	CogVLM SD
High Match (+80%)	right of, over, without	above, left, closeby	above, below, right of
Decent (70-79%)	above, left, between	below, in front of	in front of
Poor Match (¡70%)	below, behind, on top	right of, between, under	between, behind, far from

Table 5. Soft-TIFA scores on GenEval2 (800 prompts, Qwen3-VL-8B oracle). AM = arithmetic mean (atom-level); GM = geometric mean (prompt-level). †Paper-reported values for SD3.5-Large (non-Turbo).

Model	AM (%)	GM (%)
SD3.5-Large† (Kamath et al., 2025)	69.70	22.80
Standard SD3.5 (Turbo)	55.48	8.91
CLIC RedCaps (Turbo)	56.34	**10.68**
CogVLM (Turbo)	**56.52**	10.51

8.91), reflecting that the Turbo model frequently fails entire prompts rather than individual atoms. Within our Turbo variants, both CLIC variants improve over the baseline: CogVLM achieves the highest AM (56.52%) and RedCaps achieves the highest GM (10.68%), indicating that the two variants differ in whether they more reliably handle individual atoms or complete prompts.

4. Discussion & Conclusion

4.1. The Compositional Bottleneck in Diffusion Transformers

The gains in compositional accuracy, specifically an increase from 63 to 71 correct images for the phrase "to the left of" using RedCaps, and from 17 to 40 correct images for "to the right of" using CogVLM out of 100 comparisons, provide a quantitative refutation of the prevailing assumption that generative models require full-model fine-tuning to improve structural adherence. Instead, our results suggest that the Diffusion Transformer (DiT) backbone in SD3.5 possesses a latent generative "competence" that remains underutilized due to impoverished conditioning signals from standard text encoders.

While standard CLIP-L is known to exhibit "bag-of-words" behavior that prioritizes mere object presence over relational syntax, the RedCaps CLIC encoder leverages Reddit's alt-text, which is frequently authored for accessibility purposes to provide granular spatial grounding for the visually impaired. This transition from noisy, web-crawled captions to descriptive, accessibility-focused text allows the text encoder to move beyond simple object-token mapping and toward a truly relational embedding space.

The CogVLM CLIC encoder shows a complementary, but distinct, behavioral profile. In our evaluation, CogVLM delivers especially strong gains on challenging directional composition (notably "to the right of"), indicating that encoder replacement can recover spatial distinctions that the baseline text encoder under-specifies. At the representation level, CogVLM exhibits consistently larger Frobenius norms than RedCaps across attention and MLP components, with the largest relative increase in `mlp.fc2.weight` and a clear elevation in Key, Value, and Output projections. Together, these trends suggest that CogVLM supplies a stronger, more targeted compositional conditioning signal to the same SD3.5 DiT backbone, bypassing the need for computationally expensive backbone retraining.

4.2. The "Reasoning-Output Paradox" in VLMs

Our evaluation of Qwen3-VL-30B revealed a fundamental discrepancy between a model's internal reasoning traces and its final classification output, a phenomenon we term the *Reasoning-Output Paradox*.

4.2.1. Internal Representation vs. Classification Head

In multiple instances, Qwen3-VL generated a logically sound and physically accurate reasoning trace, yet concluded with an incorrect spatial answer. This suggests that the model's internal spatial awareness is significantly more advanced than its final classification head's ability to map that awareness to a rigid, categorical label.

4.2.2. Inverse Verbosity and Ambiguity

The tendency for Qwen3-VL to label standard SD3.5 outputs as "Ambiguous" more frequently in Inverse Prompting mode highlights a secondary failure mode: when the generated image itself lacks coherent spatial stratification, the VLM defaults to over-verbose descriptions that ultimately fail to trigger a definitive match with the target prompt's logic.

4.3. Deep Dive into Failure Modes: Semantic Priors vs. Structural Nouns

The exploratory phase of this project uncovered generation failures that point to a fundamental semantic hierarchy problem within the diffusion process itself.

4.3.1. The "Pizza Flyer" Phenomenon

When the model generates a literal pizza instead of a paper flyer, it visually illustrates that high-probability object tokens (e.g., "pizza") frequently override low-probability structural nouns (e.g., "flyer"). This suggests that the diffusion transformer's cross-attention layers remain heavily biased toward the most visually salient tokens in a prompt, regardless of their actual grammatical or modifying role.

4.3.2. Left-Right Asymmetry

The persistent confusion between "left" and "right", as well as "above" and "below", across all evaluated models, even after surgical encoder replacement, indicates a potential backbone-level bias. This demonstrates that even 70B+ parameter models struggle with directional absolute positioning, possibly due to the mirror-invariant properties of most underlying image-processing kernels and data augmentation strategies.

4.4. Surgical Insight: Weight Divergence in Layers

Our layer-wise analysis of the encoder replacement shows a broad increase in Frobenius norms from the early to late transformer blocks (layers 0–11), with the largest magnitudes concentrated in the final layers. This specific pattern indicates that the encoder substitution most meaningfully shifts the higher-level text representations utilized for DiT conditioning. Component-wise, CogVLM exhibits higher norms than RedCaps across both attention and MLP weights, with the largest relative increase located in `mlp.fc2.weight`. Within the attention mechanism itself, Key, Value, and Output projections are all elevated. These results are highly consistent with a modified, finer-grained token-interaction profile introduced after the encoder replacement.

4.5. Interpreting the GenEval2 TIFA Scores

The GenEval2 Soft-TIFA results reinforce, and in one case complicate, the picture drawn by our spatial-relation benchmarks. The 14 pp gap between the paper-reported SD3.5-Large score (69.7%) and our Turbo baseline (55.5%) is itself a finding: it quantifies how much compositional accuracy is sacrificed by the Turbo distillation, and suggests that our modular encoder intervention is applied to an already-constrained generative capacity.

Within our Turbo variants, both CLIC variants improve over the baseline on AM and GM, confirming that compositionally-aware text encoders provide a reliable, if incremental, grounding benefit across a diverse 800-prompt benchmark—not merely on the curated relational prompts we designed. The difference between CogVLM leading on AM while RedCaps leads on GM suggests the two variants differ in character: CogVLM more reliably handles individual visual atoms, while RedCaps is more consistent across entire prompts. One plausible explanation is that CogVLM's richer, higher-norm attention representations improve fine-grained token grounding but occasionally fail whole prompts, whereas RedCaps offers a more uniform grounding signal. This finding motivates future exploration of ensemble or skill-specific encoder selection strategies.

4.6. Limitations and Future Outlook

Despite the demonstrable success of our modular intervention, certain constraints remain:

1. **Logical Negation:** Despite some measurable improvement with the concept of "without" when using the RedCaps and CogVLM SD3.5 variants, the broader problem of strict logical negation, previously documented in earlier Stable Diffusion models, remains largely unsolved.

2. **Vague Phrasing:** Relational phrases such as "far from", "closeby", and "on" proved to be highly subjective during evaluation, leading to the lowest levels of agreement between the automated VLM predictions and our manual human annotations.

3. **Hardware Efficiency:** While we optimized the evaluation pipeline for NVIDIA A100 clusters, the sheer computational cost of large-scale VLM assessment remains a significant overhead bottleneck for rapid, iterative testing in generative model development.

4.7. Conclusion

This research establishes that compositionality is not merely an emergent property of model scale, but rather a direct consequence of training data quality and text-encoder grounding. By replacing the standard CLIP-L encoder with compositionally-aware variants like RedCaps and CogVLM, we achieved consistent improvements in spatial reasoning across both our targeted spatial-relation benchmarks and the broader GenEval2 Soft-TIFA evaluation. Ultimately, this "weight surgery" approach provides a robust, production-ready framework for enhancing the spatial reliability of text-to-image models without incurring the prohibitive computational costs associated with full-model retraining.

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5. Appendix

5.1. Classes Taken from Self-Bench for Prompt Generation

Table 6, Table 7 and Table 8 lists the unique class pairs used for generating the different prompts using the phrases “above”, “below”, “to the left of”, “to the right of”, “between”, “behind”, “closeby”, “far from”,“in front of”, “on top of”, “under”, “over”, and “without”.

Class 1	Class 2
giraffe	hair drier
broccoli	skis
book	sports ball
bench	cow
computer keyboard	skateboard
toilet	potted plant
oven	vase
bed	book
cell phone	bear
sports ball	refrigerator
baseball bat	dog
orange	traffic light
teddy bear	orange
skis	tennis racket
zebra	bicycle
sheep	computer mouse
cell phone	orange
knife	umbrella
motorcycle	carrot
traffic light	horse
surfboard	tv

Table 6. Unique class pairs used for generating the different prompts (Continued).

Class 1	Class 2
carrot	truck
giraffe	fire hydrant
elephant	toaster
bowl	dog
bench	motorcycle
frisbee	handbag
skis	cat
surfboard	spoon
bicycle	bus
car	laptop
frisbee	cat
skis	motorcycle
toilet	cat
snowboard	boat
broccoli	bird
scissors	airplane
bowl	banana
baseball glove	bicycle
microwave	skis
banana	stop sign
oven	snowboard
giraffe	tie
parking meter	toilet
banana	fork
laptop	broccoli
orange	toothbrush
baseball bat	boat
snowboard	skateboard
book	dog
chair	pizza
suitcase	train
refrigerator	sports ball
baseball bat	sandwich
potted plant	fork
clock	cow
skateboard	motorcycle
orange	bottle
backpack	wine glass
surfboard	handbag
frisbee	broccoli
cell phone	dog
chair	tennis racket
couch	tv remote
donut	knife
tv remote	laptop
traffic light	pizza
motorcycle	cat
snowboard	bottle
chair	umbrella

Table 7. Unique class pairs used for generating the different prompts.

Class 1	Class 2
fork	hot dog
clock	skateboard
potted plant	tv
laptop	computer keyboard
broccoli	dog
tv	traffic light
cell phone	parking meter
fork	snowboard
cow	carrot
truck	bus
toaster	fire hydrant
bear	pizza
pizza	frisbee
oven	donut
refrigerator	motorcycle
book	tie
hair drier	parking meter
tv	truck
teddy bear	microwave
clock	giraffe
person	refrigerator
tv	cow
microwave	cow
fire hydrant	tv remote
donut	clock
baseball glove	sandwich
hair drier	computer mouse
skateboard	bear
refrigerator	orange
clock	dining table

Table 8. Unique class pairs used for generating the different prompts (Continued).

5.2. Weight Comparison for RedCaps and CogVLM CLIC

Table 9 gives the Frobenius Norm of the difference between the original CLIP-L text encoder and the two CLIC variants layer by layer.

Table 9: Frobenius Norm of the difference between the original CLIP-L text encoder and the two CLIC variants layer by layer.

Layer Name	RedCaps CLIC	CogVLM CLIC
text_model.embeddings.position_embedding.weight	0.008173	0.010895
text_model.embeddings.token_embedding.weight	0.085015	0.108735
text_model.encoder.layers.0.layer_norm1.bias	0.00229	0.002786
text_model.encoder.layers.0.layer_norm1.weight	0.00792	0.008126
text_model.encoder.layers.0.layer_norm2.bias	0.003267	0.003584
text_model.encoder.layers.0.layer_norm2.weight	0.00894	0.00904
text_model.encoder.layers.0.mlp.fc1.bias	0.003521	0.004379
text_model.encoder.layers.0.mlp.fc1.weight	0.089096	0.111279
text_model.encoder.layers.0.mlp.fc2.bias	0.002044	0.0024
text_model.encoder.layers.0.mlp.fc2.weight	0.099352	0.121168
text_model.encoder.layers.0.self_attn.k_proj.bias	0.000566	0.000585
text_model.encoder.layers.0.self_attn.k_proj.weight	0.041081	0.049799
text_model.encoder.layers.0.self_attn.out_proj.bias	0.00209	0.002455
text_model.encoder.layers.0.self_attn.out_proj.weight	0.047979	0.059674
text_model.encoder.layers.0.self_attn.q_proj.bias	0.002145	0.002933
text_model.encoder.layers.0.self_attn.q_proj.weight	0.042259	0.04872
text_model.encoder.layers.0.self_attn.v_proj.bias	0.001954	0.002359
text_model.encoder.layers.0.self_attn.v_proj.weight	0.045594	0.056023
text_model.encoder.layers.1.layer_norm1.bias	0.002564	0.002799
text_model.encoder.layers.1.layer_norm1.weight	0.008256	0.00845
text_model.encoder.layers.1.layer_norm2.bias	0.003566	0.004009
text_model.encoder.layers.1.layer_norm2.weight	0.009192	0.009304
text_model.encoder.layers.1.mlp.fc1.bias	0.003676	0.004657
text_model.encoder.layers.1.mlp.fc1.weight	0.094367	0.118555
text_model.encoder.layers.1.mlp.fc2.bias	0.002028	0.002409
text_model.encoder.layers.1.mlp.fc2.weight	0.102252	0.126965
text_model.encoder.layers.1.self_attn.k_proj.bias	0.000493	0.000503
text_model.encoder.layers.1.self_attn.k_proj.weight	0.049575	0.063828
text_model.encoder.layers.1.self_attn.out_proj.bias	0.002076	0.00244
text_model.encoder.layers.1.self_attn.out_proj.weight	0.055471	0.069272
text_model.encoder.layers.1.self_attn.q_proj.bias	0.002261	0.002955
text_model.encoder.layers.1.self_attn.q_proj.weight	0.049195	0.062208
text_model.encoder.layers.1.self_attn.v_proj.bias	0.002267	0.002576
text_model.encoder.layers.1.self_attn.v_proj.weight	0.055425	0.069046
text_model.encoder.layers.10.layer_norm1.bias	0.002351	0.003011
text_model.encoder.layers.10.layer_norm1.weight	0.008342	0.008363
text_model.encoder.layers.10.layer_norm2.bias	0.002832	0.004221
text_model.encoder.layers.10.layer_norm2.weight	0.015948	0.016136
text_model.encoder.layers.10.mlp.fc1.bias	0.004668	0.006185
text_model.encoder.layers.10.mlp.fc1.weight	0.10911	0.12916
text_model.encoder.layers.10.mlp.fc2.bias	0.002667	0.003855
text_model.encoder.layers.10.mlp.fc2.weight	0.125197	0.159847
text_model.encoder.layers.10.self_attn.k_proj.bias	0.000441	0.000506
text_model.encoder.layers.10.self_attn.k_proj.weight	0.053107	0.062582
text_model.encoder.layers.10.self_attn.out_proj.bias	0.00243	0.003414
text_model.encoder.layers.10.self_attn.out_proj.weight	0.060872	0.06861
text_model.encoder.layers.10.self_attn.q_proj.bias	0.002373	0.003071
text_model.encoder.layers.10.self_attn.q_proj.weight	0.057276	0.069416

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Table 9: Frobenius Norm of the difference... (Continued)

Layer Name	RedCaps CLIC	CogVLM CLIC
text_model.encoder.layers.10.self_attn.v_proj.bias	0.002383	0.003037
text_model.encoder.layers.10.self_attn.v_proj.weight	0.062501	0.071082
text_model.encoder.layers.11.layer_norm1.bias	0.002951	0.003573
text_model.encoder.layers.11.layer_norm1.weight	0.008756	0.008711
text_model.encoder.layers.11.layer_norm2.bias	0.003578	0.005459
text_model.encoder.layers.11.layer_norm2.weight	0.008666	0.00925
text_model.encoder.layers.11.mlp.fc1.bias	0.006129	0.007344
text_model.encoder.layers.11.mlp.fc1.weight	0.129025	0.142604
text_model.encoder.layers.11.mlp.fc2.bias	0.003224	0.004483
text_model.encoder.layers.11.mlp.fc2.weight	0.144867	0.177204
text_model.encoder.layers.11.self_attn.k_proj.bias	0.000455	0.00053
text_model.encoder.layers.11.self_attn.k_proj.weight	0.053445	0.054786
text_model.encoder.layers.11.self_attn.out_proj.bias	0.003147	0.004476
text_model.encoder.layers.11.self_attn.out_proj.weight	0.058852	0.063449
text_model.encoder.layers.11.self_attn.q_proj.bias	0.002626	0.002632
text_model.encoder.layers.11.self_attn.q_proj.weight	0.057155	0.059257
text_model.encoder.layers.11.self_attn.v_proj.bias	0.002916	0.003503
text_model.encoder.layers.11.self_attn.v_proj.weight	0.056031	0.057719
text_model.encoder.layers.2.layer_norm1.bias	0.002412	0.00267
text_model.encoder.layers.2.layer_norm1.weight	0.008049	0.008201
text_model.encoder.layers.2.layer_norm2.bias	0.003513	0.00387
text_model.encoder.layers.2.layer_norm2.weight	0.012964	0.012956
text_model.encoder.layers.2.mlp.fc1.bias	0.003917	0.004983
text_model.encoder.layers.2.mlp.fc1.weight	0.098964	0.122465
text_model.encoder.layers.2.mlp.fc2.bias	0.001939	0.002318
text_model.encoder.layers.2.mlp.fc2.weight	0.101707	0.127526
text_model.encoder.layers.2.self_attn.k_proj.bias	0.000467	0.000496
text_model.encoder.layers.2.self_attn.k_proj.weight	0.051009	0.064013
text_model.encoder.layers.2.self_attn.out_proj.bias	0.001943	0.002303
text_model.encoder.layers.2.self_attn.out_proj.weight	0.058717	0.073337
text_model.encoder.layers.2.self_attn.q_proj.bias	0.002355	0.002809
text_model.encoder.layers.2.self_attn.q_proj.weight	0.051531	0.063047
text_model.encoder.layers.2.self_attn.v_proj.bias	0.002126	0.002457
text_model.encoder.layers.2.self_attn.v_proj.weight	0.05749	0.072235
text_model.encoder.layers.3.layer_norm1.bias	0.003124	0.003421
text_model.encoder.layers.3.layer_norm1.weight	0.008333	0.008608
text_model.encoder.layers.3.layer_norm2.bias	0.003646	0.003995
text_model.encoder.layers.3.layer_norm2.weight	0.014765	0.01487
text_model.encoder.layers.3.mlp.fc1.bias	0.004083	0.005219
text_model.encoder.layers.3.mlp.fc1.weight	0.103157	0.12407
text_model.encoder.layers.3.mlp.fc2.bias	0.001906	0.00239
text_model.encoder.layers.3.mlp.fc2.weight	0.107144	0.132704
text_model.encoder.layers.3.self_attn.k_proj.bias	0.000454	0.000484
text_model.encoder.layers.3.self_attn.k_proj.weight	0.057809	0.069354
text_model.encoder.layers.3.self_attn.out_proj.bias	0.001951	0.002359
text_model.encoder.layers.3.self_attn.out_proj.weight	0.062525	0.077634
text_model.encoder.layers.3.self_attn.q_proj.bias	0.002541	0.003468
text_model.encoder.layers.3.self_attn.q_proj.weight	0.058596	0.068339
text_model.encoder.layers.3.self_attn.v_proj.bias	0.002037	0.002453

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Table 9: Frobenius Norm of the difference... (Continued)

Layer Name	RedCaps CLIC	CogVLM CLIC
text_model.encoder.layers.3.self_attn.v_proj.weight	0.061898	0.076816
text_model.encoder.layers.4.layer_norm1.bias	0.002604	0.003014
text_model.encoder.layers.4.layer_norm1.weight	0.008217	0.008424
text_model.encoder.layers.4.layer_norm2.bias	0.003601	0.004041
text_model.encoder.layers.4.layer_norm2.weight	0.015312	0.015459
text_model.encoder.layers.4.mlp.fc1.bias	0.004018	0.005087
text_model.encoder.layers.4.mlp.fc1.weight	0.105542	0.126395
text_model.encoder.layers.4.mlp.fc2.bias	0.001791	0.002343
text_model.encoder.layers.4.mlp.fc2.weight	0.105839	0.133286
text_model.encoder.layers.4.self_attn.k_proj.bias	0.000382	0.000455
text_model.encoder.layers.4.self_attn.k_proj.weight	0.056201	0.069655
text_model.encoder.layers.4.self_attn.out_proj.bias	0.001882	0.002335
text_model.encoder.layers.4.self_attn.out_proj.weight	0.063063	0.080001
text_model.encoder.layers.4.self_attn.q_proj.bias	0.002408	0.003384
text_model.encoder.layers.4.self_attn.q_proj.weight	0.059726	0.073507
text_model.encoder.layers.4.self_attn.v_proj.bias	0.001827	0.002351
text_model.encoder.layers.4.self_attn.v_proj.weight	0.06099	0.076788
text_model.encoder.layers.5.layer_norm1.bias	0.002729	0.00334
text_model.encoder.layers.5.layer_norm1.weight	0.00811	0.008459
text_model.encoder.layers.5.layer_norm2.bias	0.003539	0.003892
text_model.encoder.layers.5.layer_norm2.weight	0.015581	0.015487
text_model.encoder.layers.5.mlp.fc1.bias	0.004108	0.005241
text_model.encoder.layers.5.mlp.fc1.weight	0.109652	0.130334
text_model.encoder.layers.5.mlp.fc2.bias	0.001734	0.002295
text_model.encoder.layers.5.mlp.fc2.weight	0.105196	0.133991
text_model.encoder.layers.5.self_attn.k_proj.bias	0.000397	0.000479
text_model.encoder.layers.5.self_attn.k_proj.weight	0.055773	0.068963
text_model.encoder.layers.5.self_attn.out_proj.bias	0.001737	0.00235
text_model.encoder.layers.5.self_attn.out_proj.weight	0.060264	0.080776
text_model.encoder.layers.5.self_attn.q_proj.bias	0.002164	0.003065
text_model.encoder.layers.5.self_attn.q_proj.weight	0.056652	0.069674
text_model.encoder.layers.5.self_attn.v_proj.bias	0.001693	0.00237
text_model.encoder.layers.5.self_attn.v_proj.weight	0.057974	0.077634
text_model.encoder.layers.6.layer_norm1.bias	0.002379	0.002759
text_model.encoder.layers.6.layer_norm1.weight	0.008284	0.008269
text_model.encoder.layers.6.layer_norm2.bias	0.00386	0.00423
text_model.encoder.layers.6.layer_norm2.weight	0.015301	0.015312
text_model.encoder.layers.6.mlp.fc1.bias	0.004155	0.005143
text_model.encoder.layers.6.mlp.fc1.weight	0.108709	0.130725
text_model.encoder.layers.6.mlp.fc2.bias	0.001783	0.002332
text_model.encoder.layers.6.mlp.fc2.weight	0.105817	0.135144
text_model.encoder.layers.6.self_attn.k_proj.bias	0.00046	0.000528
text_model.encoder.layers.6.self_attn.k_proj.weight	0.056081	0.068587
text_model.encoder.layers.6.self_attn.out_proj.bias	0.001763	0.002342
text_model.encoder.layers.6.self_attn.out_proj.weight	0.061265	0.078309
text_model.encoder.layers.6.self_attn.q_proj.bias	0.002066	0.002653
text_model.encoder.layers.6.self_attn.q_proj.weight	0.058913	0.069418
text_model.encoder.layers.6.self_attn.v_proj.bias	0.001692	0.002261
text_model.encoder.layers.6.self_attn.v_proj.weight	0.059581	0.076724

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Table 9: Frobenius Norm of the difference... (Continued)

Layer Name	RedCaps CLIC	CogVLM CLIC
text_model.encoder.layers.7.layer_norm1.bias	0.002778	0.00314
text_model.encoder.layers.7.layer_norm1.weight	0.008431	0.00839
text_model.encoder.layers.7.layer_norm2.bias	0.00349	0.003684
text_model.encoder.layers.7.layer_norm2.weight	0.015476	0.01543
text_model.encoder.layers.7.mlp.fc1.bias	0.004086	0.005151
text_model.encoder.layers.7.mlp.fc1.weight	0.10613	0.128304
text_model.encoder.layers.7.mlp.fc2.bias	0.001893	0.002423
text_model.encoder.layers.7.mlp.fc2.weight	0.10737	0.134731
text_model.encoder.layers.7.self_attn.k_proj.bias	0.000425	0.000508
text_model.encoder.layers.7.self_attn.k_proj.weight	0.053459	0.064851
text_model.encoder.layers.7.self_attn.out_proj.bias	0.001848	0.002337
text_model.encoder.layers.7.self_attn.out_proj.weight	0.060451	0.076152
text_model.encoder.layers.7.self_attn.q_proj.bias	0.002237	0.002805
text_model.encoder.layers.7.self_attn.q_proj.weight	0.056506	0.068229
text_model.encoder.layers.7.self_attn.v_proj.bias	0.001787	0.002218
text_model.encoder.layers.7.self_attn.v_proj.weight	0.058207	0.075241
text_model.encoder.layers.8.layer_norm1.bias	0.002233	0.002667
text_model.encoder.layers.8.layer_norm1.weight	0.008338	0.008513
text_model.encoder.layers.8.layer_norm2.bias	0.003256	0.003737
text_model.encoder.layers.8.layer_norm2.weight	0.015716	0.015836
text_model.encoder.layers.8.mlp.fc1.bias	0.004069	0.005309
text_model.encoder.layers.8.mlp.fc1.weight	0.106005	0.127332
text_model.encoder.layers.8.mlp.fc2.bias	0.00206	0.002734
text_model.encoder.layers.8.mlp.fc2.weight	0.112115	0.141943
text_model.encoder.layers.8.self_attn.k_proj.bias	0.000412	0.000481
text_model.encoder.layers.8.self_attn.k_proj.weight	0.051519	0.067374
text_model.encoder.layers.8.self_attn.out_proj.bias	0.002003	0.002552
text_model.encoder.layers.8.self_attn.out_proj.weight	0.059103	0.071199
text_model.encoder.layers.8.self_attn.q_proj.bias	0.002087	0.002898
text_model.encoder.layers.8.self_attn.q_proj.weight	0.053	0.069356
text_model.encoder.layers.8.self_attn.v_proj.bias	0.001918	0.002482
text_model.encoder.layers.8.self_attn.v_proj.weight	0.058047	0.071918
text_model.encoder.layers.9.layer_norm1.bias	0.003024	0.003472
text_model.encoder.layers.9.layer_norm1.weight	0.007936	0.008213
text_model.encoder.layers.9.layer_norm2.bias	0.003413	0.003914
text_model.encoder.layers.9.layer_norm2.weight	0.015666	0.015855
text_model.encoder.layers.9.mlp.fc1.bias	0.004386	0.005424
text_model.encoder.layers.9.mlp.fc1.weight	0.108228	0.126033
text_model.encoder.layers.9.mlp.fc2.bias	0.002245	0.003108
text_model.encoder.layers.9.mlp.fc2.weight	0.116033	0.146626
text_model.encoder.layers.9.self_attn.k_proj.bias	0.000478	0.000577
text_model.encoder.layers.9.self_attn.k_proj.weight	0.05651	0.062531
text_model.encoder.layers.9.self_attn.out_proj.bias	0.002153	0.002809
text_model.encoder.layers.9.self_attn.out_proj.weight	0.064986	0.072812
text_model.encoder.layers.9.self_attn.q_proj.bias	0.002454	0.002814
text_model.encoder.layers.9.self_attn.q_proj.weight	0.058875	0.066071
text_model.encoder.layers.9.self_attn.v_proj.bias	0.002173	0.002791
text_model.encoder.layers.9.self_attn.v_proj.weight	0.065208	0.074562
text_model.final_layer_norm.bias	0.003565	0.005115

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Table 9: Frobenius Norm of the difference... (Continued)

Layer Name	RedCaps CLIC	CogVLM CLIC
text_model.final_layer_norm.weight	0.00607	0.006321
text_projection.weight	29.918453	29.918283